



1. SMIC PDK installation Procedure and Directory

SMIC PDK is distributed in compressed tar format,
SPDK18MSE_1833_OA(_CDB)_CDS_Vx.x_x.tar.gz

At this stage, you will know how to install SMIC PDK with SMIC PDK installer.

1. Make a new directory, <PDK_Path>, copy the PDK package,
2. Use the command **gtar zxvf SPDK18MSE_1833_OA(_CDB)_CDS_Vx.x_x.tar.gz** to decompress the file.

3. Completing the 1 and 2 , we could summarize the PDK directory as the following



README.INSTALL: Installation script note



SMIC_PDK_install: PDK installation script



smic18mse_1833_oa(cdb)_cds_vx.x_x: It is a directory, and it

includes all the PDK related files. Strongly do not suggest modifying any information under this directory.

4. Prerequisite

You need to source ICFB/Virtuoso environment before starting to install PDK. And you need to install to PERL. The recommend tool version is as below:

IC5141ISR150 (CDB PDK)

Virtuoso DFII version IC6.1.6-64b.500.3 (OA PDK)

Virtuoso schematic editor L,

Virtuoso Layout Suite L & XL,

Virtuoso ADE Environment,

Spectre MMSIM12.11.188

Hspice 2010.12-SP2,

Calibre 2013.4_37.29

PERL 5.8 or above

5. Execute "SMIC_PDK_install" command



./ SMIC_PDK_install

6. Option choices:

Available options are as below, please select your prefer option.

Please choose 1.8V/3.3V process or pure 3.3V process for installation:

1) 1.8V/3.3V

2) 3.3V

Enter 1 or 2: 1

Please choose the number of top metal for installation:

1) 1 Top Metal

Enter 1: 1

Please choose the type of 1 Top metal for installation:

1) 9K top metal layer, thickness 9kA (TMa1)

2) 22K top metal layer, thickness 22kA (TMa2)

3) 34K top metal layer, thickness 34kA (TMa3)

4) 40K top metal layer, thickness 40kA (TMa4)

Enter 1, 2, 3 or 4: 1

Please choose the number of inter metal layers:

1) 2 inter metals

2) 3 inter metals

3) 4 inter metals

4) 5 inter metals

Enter 1, 2, 3 or 4: 4

Please choose the type of MIM for installation:

1) MIM1.0fF, Cspec=1fF

2) MIM2.0fF, Cspec=2fF

Enter 1 or 2 : 1

The PDK metal option is: 1P6M_5la_1TMa1_MIM10

Do you want to install yes/no?[yes|no]:yes

Please confirm your choice in final.

2. SMIC PDK Directory

After you have finished PDK option, a new PDK directory will be created. For example:

SMIC Confidential

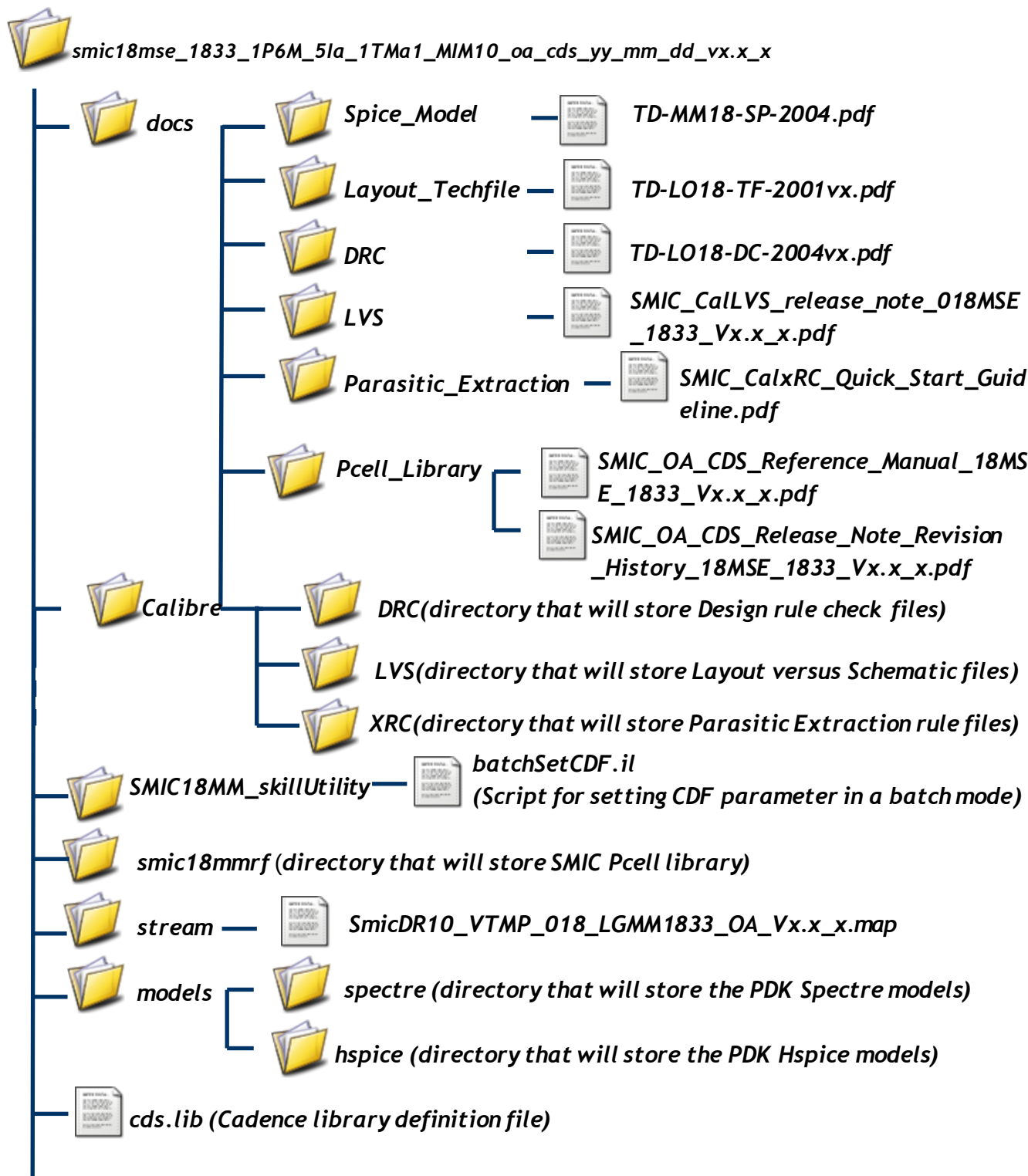
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smic18mse_1833_1P6M_5la_1TMa1_MIM10_oa_cds_yy_mm_dd_vx.x_x. We could summarize the PDK directory structures as followings.





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-  *display.drf (Cadence Display Resources)*
-  *techfile.tf (ASCII version of technology file)*



3. Create Schematic with Schematic Editor

3.1 Schematic design

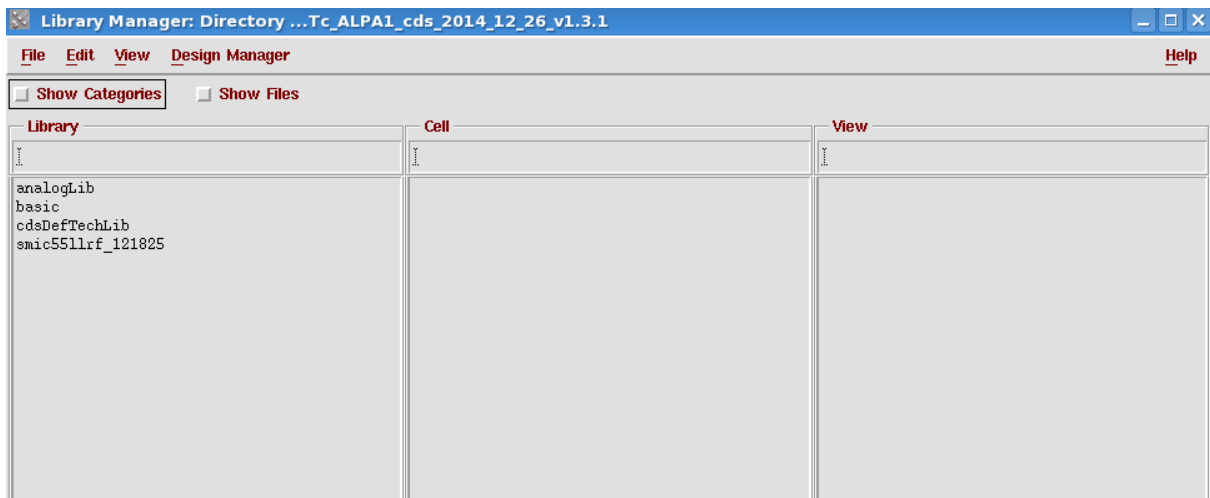
After you have installed your PDK, you can start to work with SMIC Cadence PDK. The following is an “Inventor” example which demonstrates the Cadence analog design flow based on 55nm PDK.

Type **icfb&** in the command line, and then you can get the **Command Interface Window (CIW)**.

Open Library Manager

Select **Tools -> Library Manager**

LIBRARY MANAGER



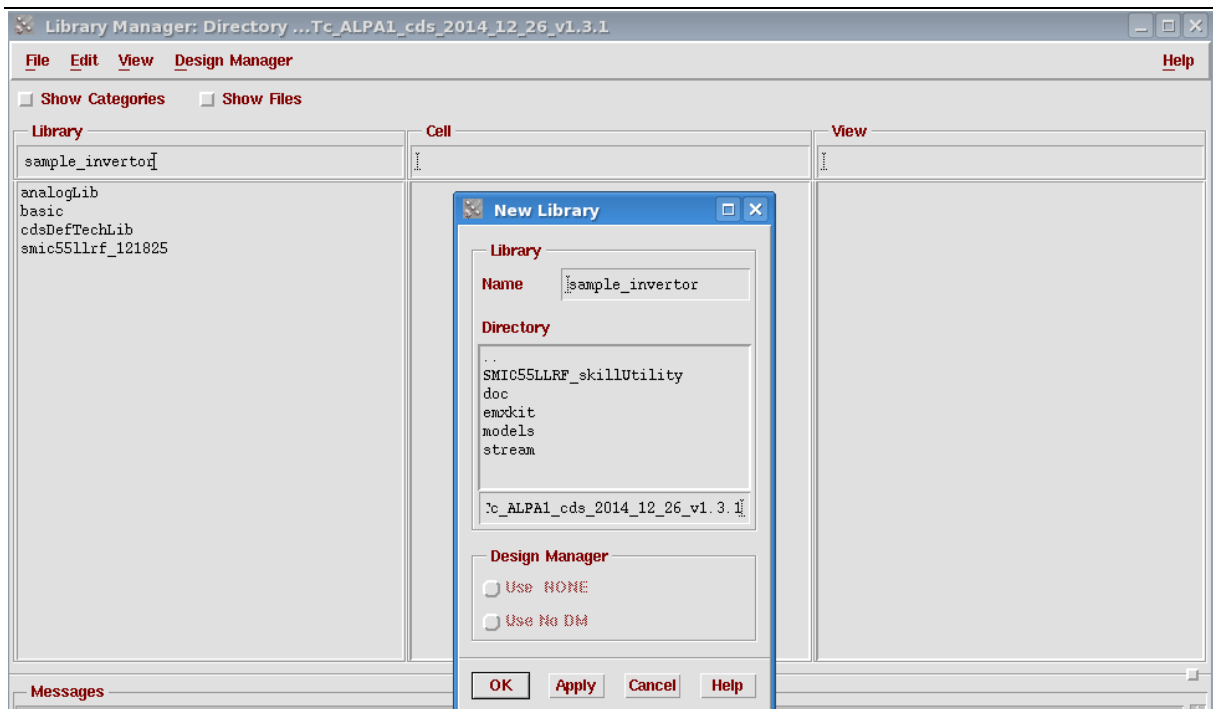
Creating a New Library

1. Select **File -> New->Library** in Library Manager.

NEW LIBRARY

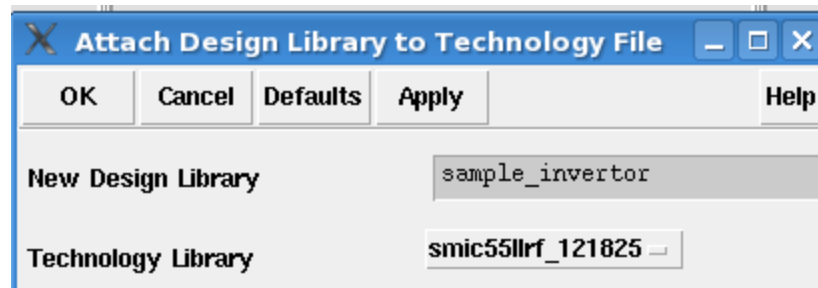


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2. Type in a library name, and choose '**Attach to an existing techfile**' and choose the **TechnologyLibrary** name as "smic55llrf_121825".

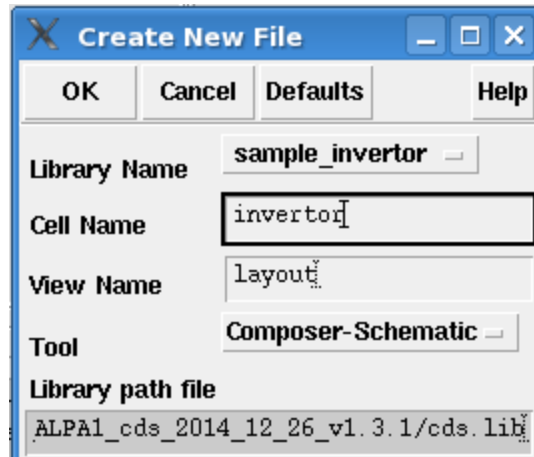
ATTACH DESIGN LIBRARY TO TECHNOLOGY FILE



Creating a Schematic view

1. Select **File -> new -> Cell view** in Library Manager, and choose the created library to create a schematic view. Type in **Cell** and **View** name and choose **Schematic** as the Type.

CREATE NEW SCHEMATIC



Placing Instances on Schematic

Add transistors or other device from PDK library into the schematic.

From the Virtuoso Schematic Editor window, select **Add->Instance**. This brings out the **Add Instance** form. Click on **Browse**, the **Library Browser** window will appear. To select device from PDK, you should have the **Add Instance** window.

ADD INSTANCE



Add Instance

Hide Cancel Defaults Help

Library: smic5511rf_121825 Browse

Cell: p1211

View: symbol

Names:

Array: Rows: 1 Columns: 1

Rotate Sideways Upside Down

Model Name: p1211

Multiplier: 1

Length: 60n

Total Width: 240n

Finger Width: 240n

Fingers: 1

Threshold: 120n

Apply Threshold: ☐

Gate Connection: ☒ None ☐ Top ☐ Bottom

Gate Direction: ☒ Mid ☐ Left ☐ Right

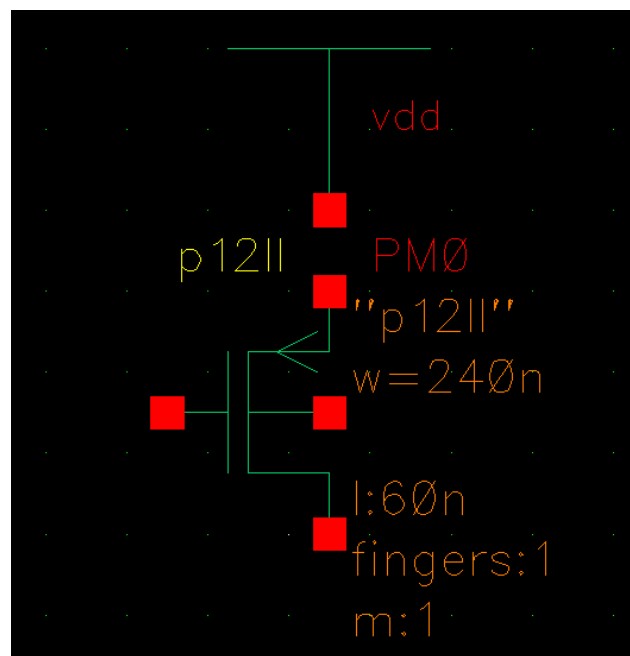
Use DFM Rules: ☒ DFM

CT enc by S/D AA (horizontal): 15n

multi CTs enc by S/D AA (vertical): 75n

Add ground, power supply with symbol in “analogLib”.

ADD POWER SUPPLY



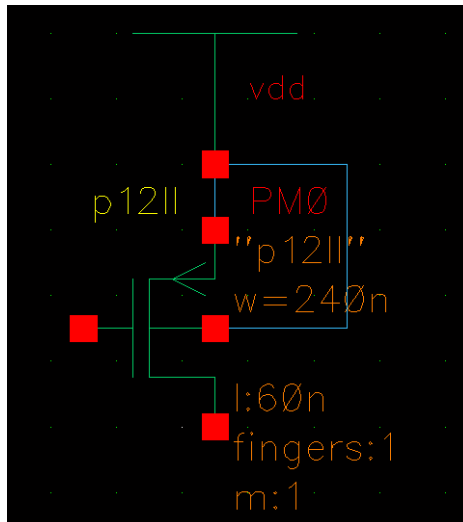
Add wire.



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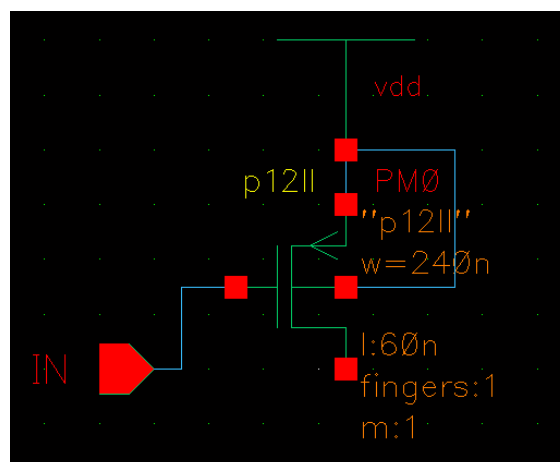
After the instances are created and placed, then you need to connect them by wires. From the Virtuoso Schematic Editor window, select **Add->Wire**.

ADD WIRE

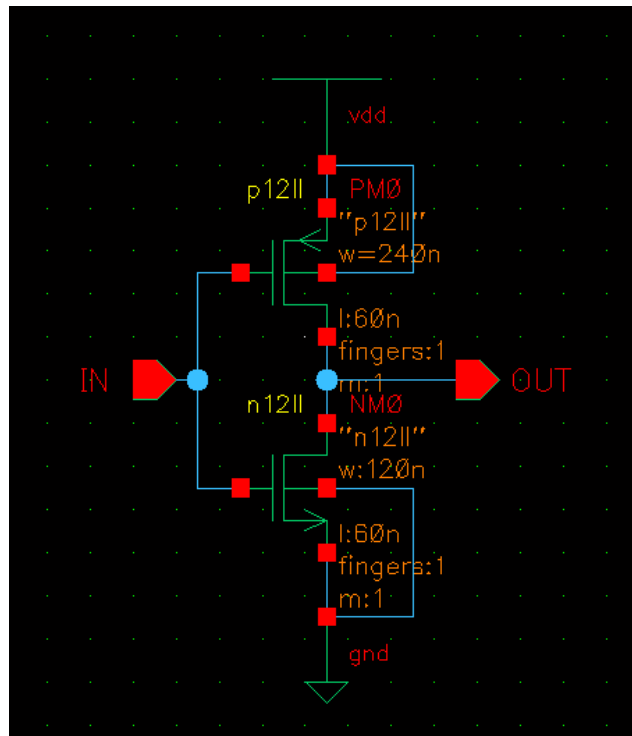


Then you need to create pins for input and output nodes, select **Add->Pin**.

ADD PIN



Then the inverter schematic is created.



Generating Symbol View

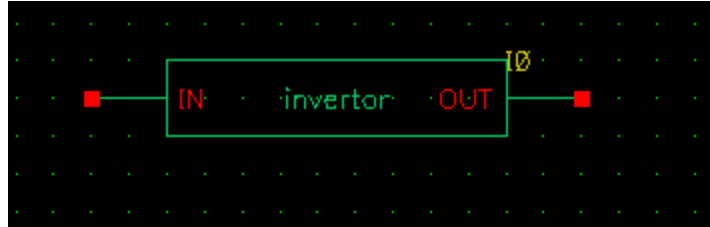
In schematic view, select **Design->Create Cellview->From Cellview**. Type in the name for Cell and View and choose schematicSymbol.

SYMBOL GENERATION



Import a symbol

Select **Add->Instance** and you could instance the symbol you just created.



3.2 Pre-simulation in Spectre with Virtuoso Analog Design Environment

This stage introduces how to simulate your circuit with Virtuoso Analog Design Environment (ADE).

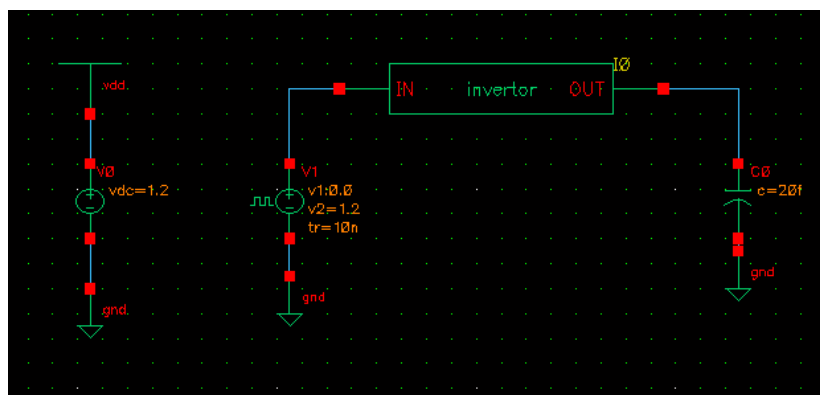
Create a Test Bench

To simulate your design, you need to create a test bench with cell named as 'testbench_invertor' with the symbol you create before, power supply, input source, and a ground are included in this schematic.

We will start to simulate the transient response.

Firstly create new schematic view 'testbench_invertor' and instance the symbol 'inverter'. Add the input source and power supply.

TEST BENCH SCHEMATIC



POWER SUPPLY



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Property	Value	Display
Library Name	analogLib	off ☐
Cell Name	vdc	off ☐
View Name	symbol	off ☐
Instance Name	V0	off ☐

User Property	Master Value	Local Value	Display
lvignore	TRUE		off ☐

CDF Parameter	Value	Display
AC magnitude		off ☐
AC phase		off ☐
DC voltage	1.2 V	off ☐
Noise file name		off ☐

INPUT SOURCE

Property	Value	Display
Library Name	analogLib	off ☐
Cell Name	vpulse	off ☐
View Name	symbol	off ☐
Instance Name	V1	off ☐

User Property	Master Value	Local Value	Display
lvignore	TRUE		off ☐

CDF Parameter	Value	Display
AC magnitude		off ☐
AC phase		off ☐
DC voltage		off ☐
Voltage 1	0.0 V	off ☐
Voltage 2	1.2 V	off ☐
Delay time	10n	off ☐
Rise time	10n	off ☐
Fall time	10n	off ☐
Pulse width	10n	off ☐
Period	50n	off ☐

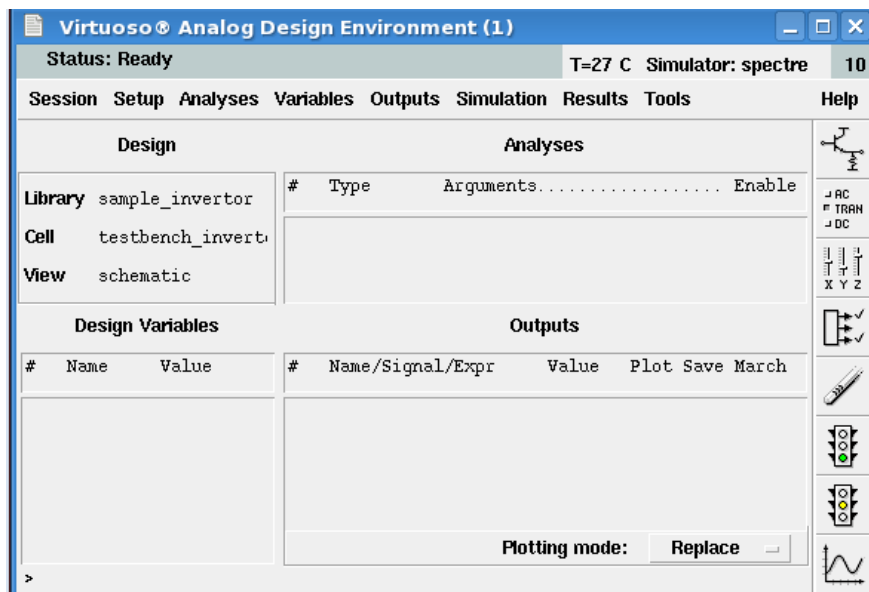
Simulate the Test Bench with SMIC Models

When simulate your design 'testbench_invertor', please follow the below steps.

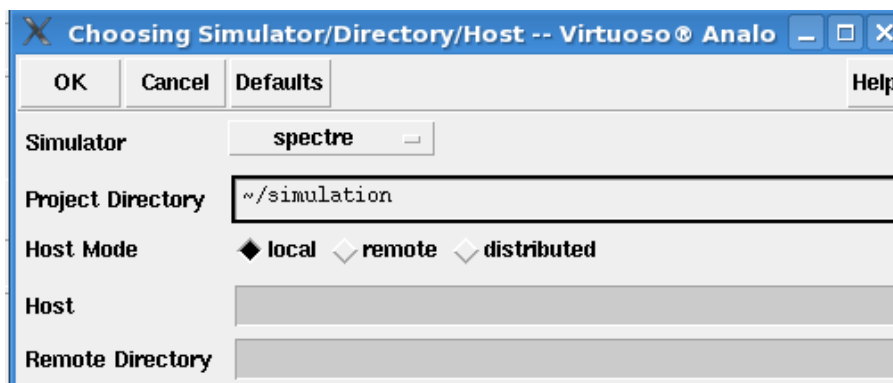
In the Virtuoso Schematic Editor window, select **Tool->Analog Environment**.



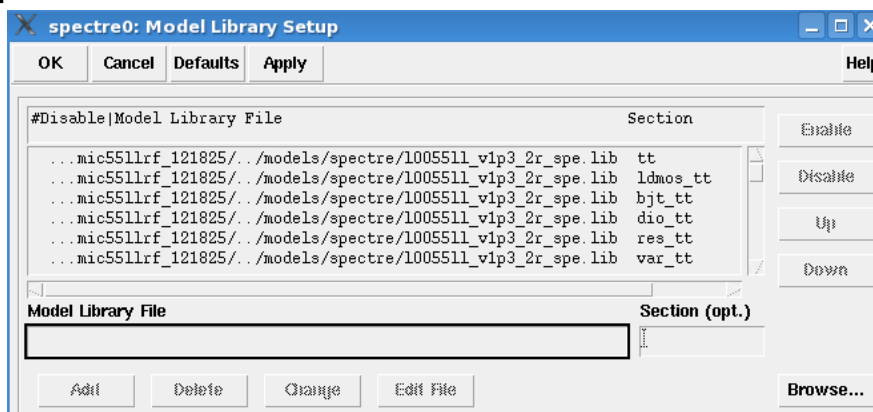
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Select Simulator as 'spectre'.



Select Setup- >Model Libraries.

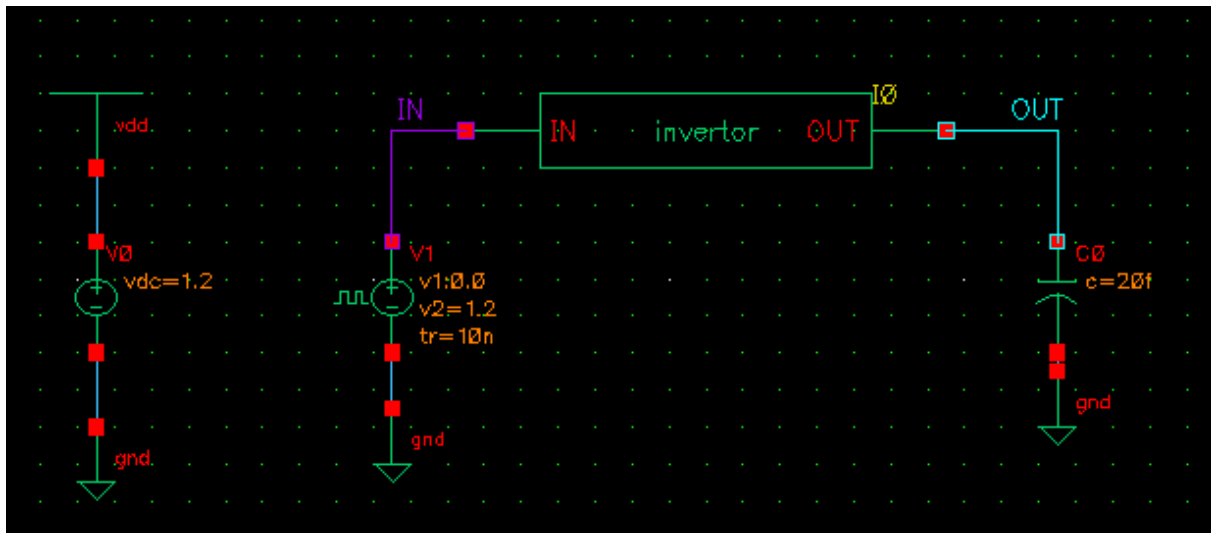




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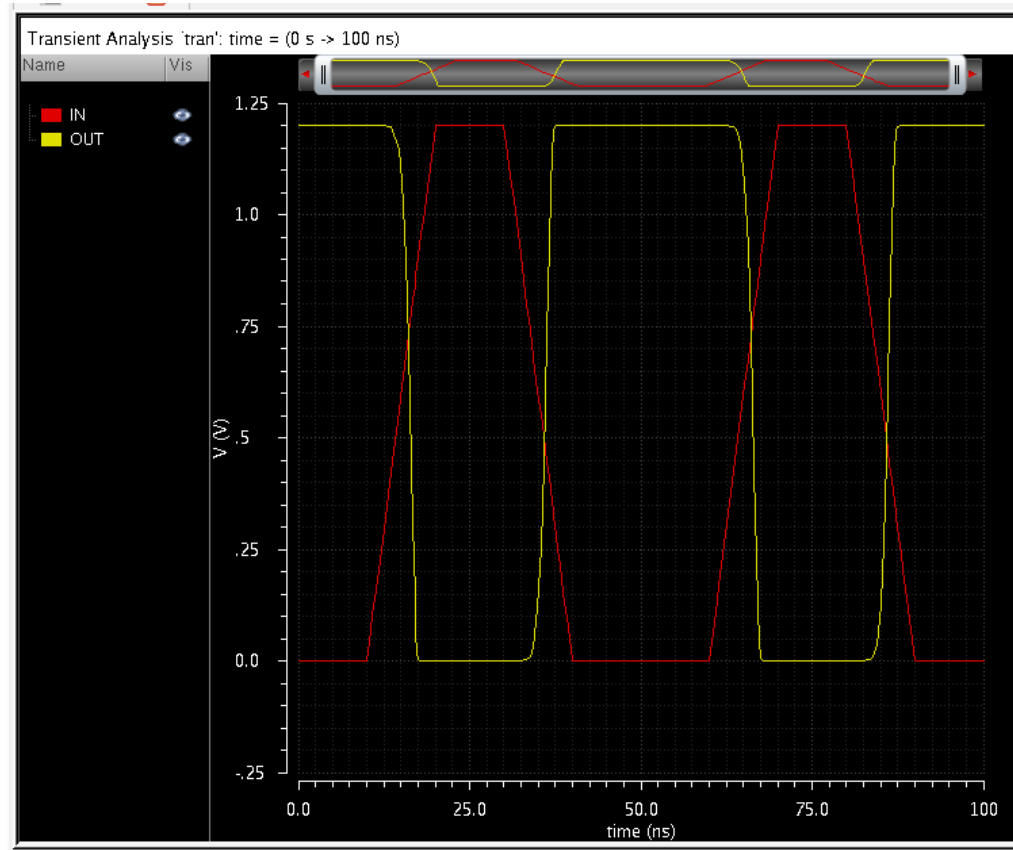
Select **Analyses->Choose**, and then fill out the form as below:

Select **Outputs->To Be Plotted->Select On Schematic**. Then, go back to the schematic and select the 'IN' and 'OUT' wire.





Select **Simulation->Run**. Then, the plot of the simulation will appear in Waveform Window.



4. Create Layout with Layout Editor XL

After a schematic is completed, the next step is to create layout view by using the Virtuoso Layout Suite. We will use PCells which are provided in SMIC PDK to create an layout. The CADENCE Virtuoso Layout XL design flow will be implemented.

The schematic-driven layout in the Virtuoso Layout XL can be used to auto create a correct design layout.



In the Schematic Editor window, select **Tools->Design Synthesis->Layout XL**

In the Virtuoso Layout XL window, select **Design ->Gen From Source**

LAYOUT GENERATION OPTIONS

Layout Generation

Generate: ☒ I/O Pins ☒ Instances ☒ Boundary
☐ Transistor Chaining ☐ Transistor Folding ☐ Preserve Mappings

I/O Pins

Apply Pin Type Layer / Master Width Height Num Create

Defaults: Geometric M1 dg 0.09 0.09 1

Select: Number Selected: 0 Add a Pin

Term Name	Net Name	Pin Type	Layer / Master	Width	Height	Num	Create
"IN"	"IN"	"Geometric"	("M1" "drawing")	0.09	0.09	1	
"OUT"	"OUT"	"Geometric"	("M1" "drawing")	0.09	0.09	1	
"gnd!"	"gnd!"	"Geometric"	("M1" "drawing")	0.09	0.09	1	
"vdd!"	"vdd!"	"Geometric"	("M1" "drawing")	0.09	0.09	1	

Update Pin Type Layer / Master Width Height Num Create

AS IS Geometric M1 dg 0.09 0.09 1

Pin Label Shape: ☐ Label ☐ Text Display ☒ None Pin Label Options...

Boundary

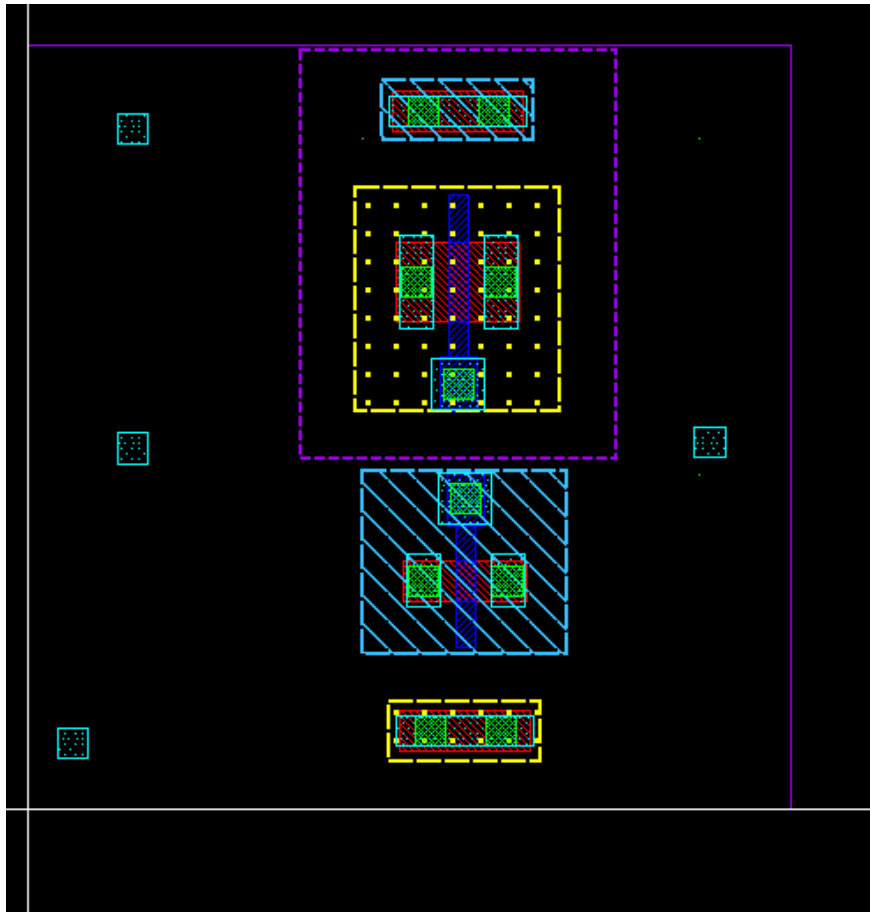
Layer: prBndry dg Left: Bottom:

Shape: Rectangle

Boundary Area Estimation

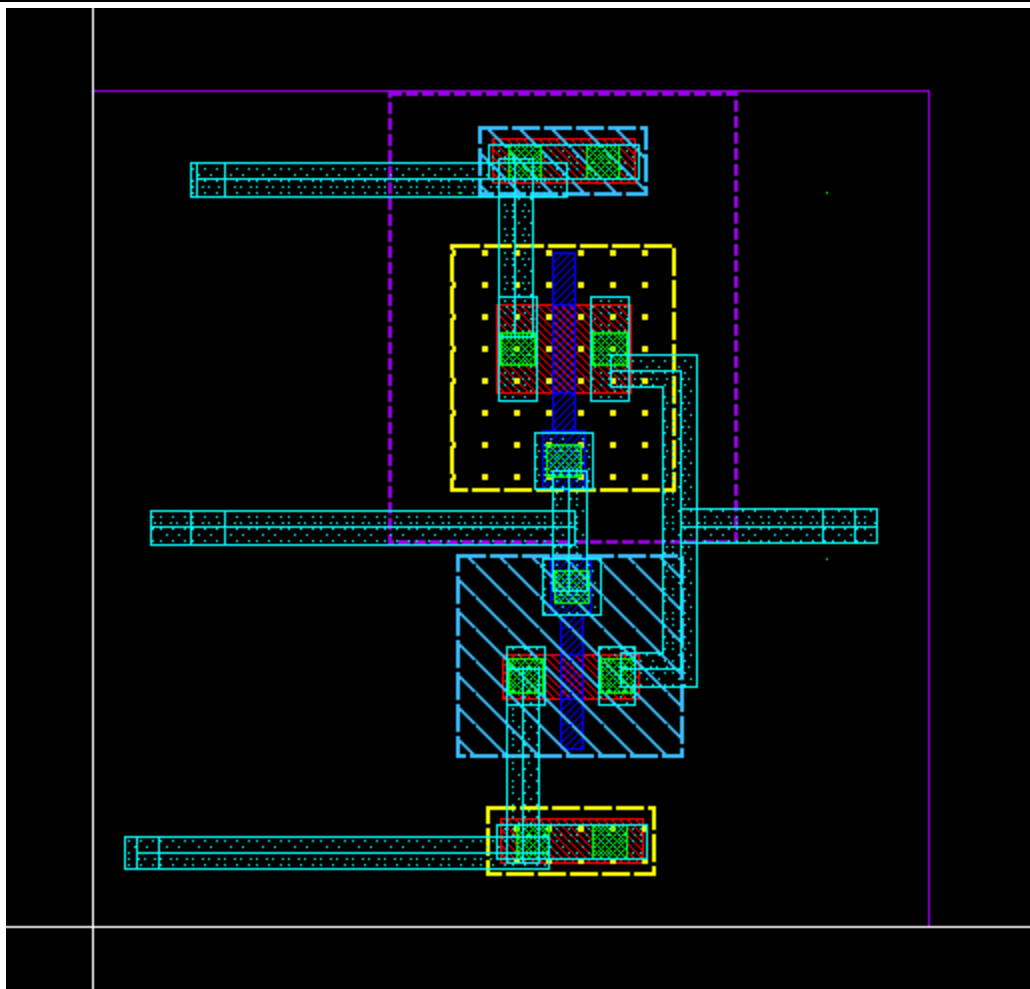
Utilization (%) 25 Aspect Ratio (W/H) 1

LAYOUT VIEW



You can use the **VCAR** and the **icc.rules** which is provided in Cadence PDK for auto routing.

LAYOUT VIEW AFTER ROUTING

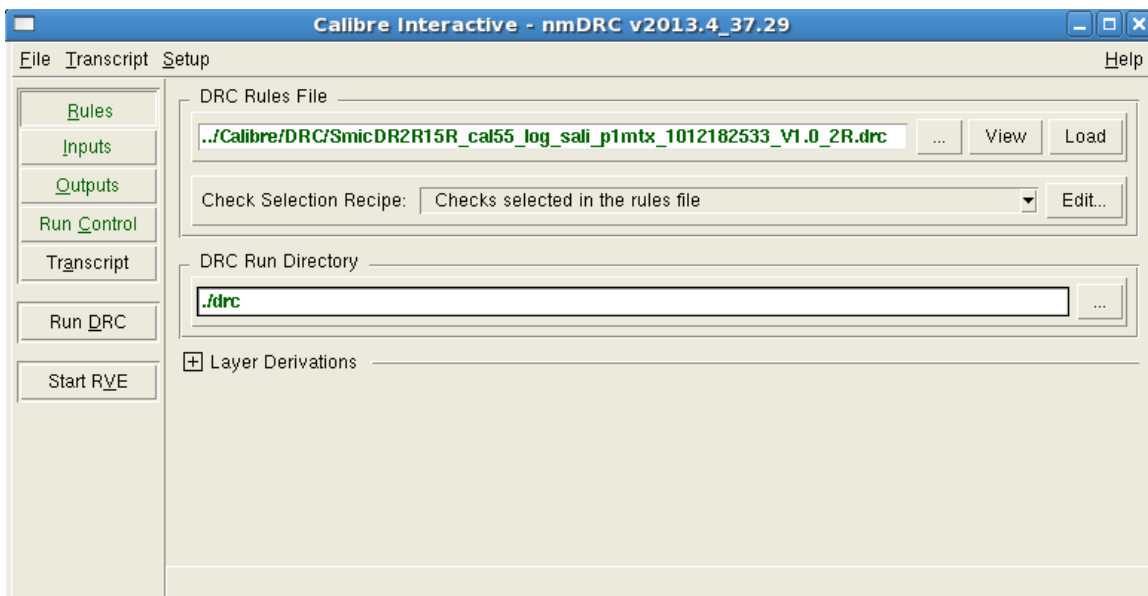




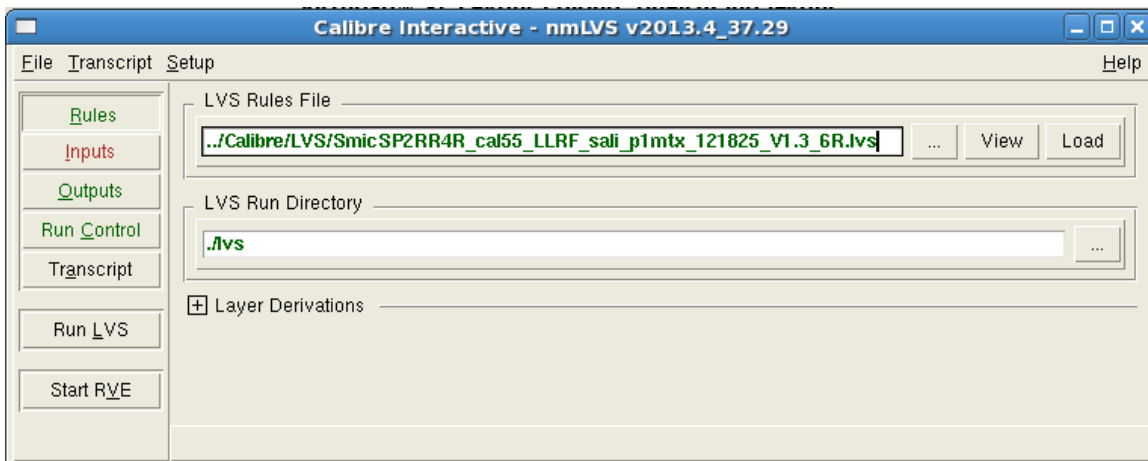
5. Physical Verification

As the layout is completed, we are going to run the physical verification flow to ensure that the layout contains no DRC violations and each device in the layout completely matches with its corresponding component in the schematic.

RUN DRC



RUN LVS



RUN XRC (please refer quick-start_on_smic-calibrexrc.pdf)



6. Post Layout Simulation

After the layout is both DRC and LVS clean, the next step is to perform the RC extraction.